April 9, 2022

Dr. Stephanie Johnson
U.S. Department of Energy
Office of Energy Efficiency and Renewable Energy
Building Technologies Office EE-2J
1000 Independence Avenue, SW
Washington, DC 20585-0121
Email: AirCleaners2021STD0035@ee.doe.gov
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Docket: EERE-2021-BT-STD-0035, EERE-2021-BT-TP-0036, EERE-2021-BT-DET-0022

***Re: Comments to Docket Number EERE-2021-BT-STD-0035***

Dear Dr. Johnson:

Molekule, Inc. ("**Molekule**" or the "**Company**") produces home and commercial air purification systems based on its founders' pathbreaking efforts to reduce exposure to airborne impurities and pathogens. Molekule, based in San Francisco, California, was founded in 2014 by American scientist Dr. Yogi Goswami, his son Dilip Goswami, and his daughter Jaya Rao.

Molekule's Photoelectric Chemical Oxidization ("**PECO**") technology is unique and distinct from traditional filtration-only air purifiers, utilizing free radicals to oxidize and destroy pollutants as small as 0.1 nanometers across. Molekule's Air Mini+ and Air Pro products have been cleared by the U.S. Food and Drug Administration (the **"FDA"**) as Class II medical devices for the destruction of airborne bacteria and viruses for medical purposes (the Air Pro and Air Pro RX devices have also been cleared for the destruction of mold) (K200500, K202339, K211194).

Molekule's novel PECO technology goes beyond traditional filtration-only air purifiers. PECO addresses a wide range of pollutants, including many organic pollutants smaller than those tested using only the traditional filter standard focused on particulate matter. In order to address this wide range of pollutants, Molekule's devices depend on a combination of an air circulating fan, a catalytic filter and an internal light source utilized to activate the catalyst and enable the destruction of microscopic pollutants. Accordingly, the energy requirements for Molekule devices needed to unlock the benefits of PECO technology are fundamentally different from those of traditional filtration-only air purifiers.

In order to make certain that Molekule's PECO air purifiers continue to meet minimum FDA-reviewed performance requirements, and to promote continuing innovation in the air purifier space, including the development of novel technologies such as PECO, Molekule respectfully requests that the Department of Energy (the "**DOE**") ensure that any federal energy efficiency requirements for air purifiers appropriately account for technologies that may provide benefits beyond traditional air filtration-only devices. As PECO shows, such technologies can have differing energy requirements to be able to target potentially harmful pollutants such as viruses and organic chemicals, in addition to the particulate matter that is the historical focus of traditional filtration. For example, energy standards based solely on traditional Clean Air Delivery Rate ("**CADR**")-based testing methods that only focus on particulate capture are likely to hinder the development and distribution of innovative devices with capabilities of addressing the wide range of harmful pollutants found in our indoor air.

Molekule specifically requests that any test procedures and standards applicable to air purifiers include the following provisions, each of which is explained in more detail below:

1. **FDA-Cleared Device Exemption** – An exemption from federal energy efficiency requirements for air purifiers that have U.S. Food and Drug Administration (the "**FDA**") clearance or approval as a medical device in accordance with section 360c of title 21 of the U.S. Code.
2. **Combined Technology Devices Adjustment** – Adjusted requirements for any air purifiers utilizing combined technologies such as a fan and ultraviolet light sources that are intended to capture and destroy a wide range of potentially harmful pollutants. Molekule proposes an additional 15% energy allowance for devices featuring technology with capabilities outside of the scope of AHAM AC-1 and its smoke, dust, pollen tests.
3. **Automatic/Standby Mode Adjustment** – Adjusted requirements to account for the use of air purifiers in "automatic" or "standby" mode. Molekule proposes that the DOE include a credit mechanism for such devices. Due to the complexity of this area, we believe that further input and analysis will be required from manufacturers and the DOE to establish appropriate standards and Molekule recommends a delayed implementation of energy efficiency requirements for devices with such "automatic" or "standby" functionality until such standards are determined.

***Background on Molekule and PECO Technology***

Molekule currently manufactures two devices: the Molekule Air Mini+ and the Molekule Air Pro.[1] Both devices utilize a multilayer PECO filter to capture pollutants and destroy them while they move through the filter to the catalytic surface. PECO works by shining UV-A light on the catalytic surface of the PECO filter. The UV-A light provides enough energy to excite electrons in the valence band of the catalyst to the conduction band, leaving an electron hole, which attracts electrons from negatively charged hydroxyl ions (OH-) that are ubiquitous in the air. The donation of an electron from a hydroxyl ion generates the highly reactive but short-lived hydroxyl radical (OH*). As the air passes through the catalytic filter, the contaminants and

[1] The Company previously produced and has limited inventories of the Molekule Air, Molekule Air Mini and Molekule Air Pro RX devices.

other particles get trapped on the catalytic filter. When UV light rays shine on the catalytic surface, the hydroxyl radicals that are created combine with airborne microbiological contaminants, such as bacteria, viruses, and mold spores entrained onto the filter fibers or surface. Once combined, a chemical reaction takes place, destroying the contaminants. All hydroxyl radicals are consumed in the reactions on the surface of the filter and do not leave the device.

Molekule has conducted numerous tests, including research done at third party institutions, to demonstrate the effectiveness of its devices in capturing and destroying pollutants in the air, including destroying airborne viruses, bacteria, mold and volatile organic chemicals. Results are available on www.molekule.com/papers.

***FDA 510(k) Cleared Device Exemption***

Molekule's Air Pro and Air Mini + devices have been cleared by the FDA as Class II medical devices for the destruction of viruses and bacteria for medical purposes (the Air Pro was also cleared for the destruction of mold). The clearances allow medical professionals to use these devices in medical settings to purify the air for these contaminants. The removal and destruction of airborne microbes is a key benefit in medical settings but is not measured by CADR tests for particulate matter.

Significant expenses and resources are required to apply for and obtain 510(k) clearances from the FDA. The 510(k) premarket notifications for Molekule's Air Pro and Air Mini + devices included verification and validation testing demonstrating that the devices safely and effectively destroy viruses and bacteria, and that the devices met FDA-required electrical safety and electromagnetic compatibility standards.

Any modifications to the design of cleared devices, including modifications that may be required to meet new energy efficiency requirements, must follow stringent FDA requirements that can require significant time, effort and expense to implement. In this case, imposition of new energy efficiency requirements may require a redesign of the FDA-cleared devices, which may render the verification and validation testing included in Molekule's notifications inapplicable. Molekule could be required by FDA to repeat the testing included in its original submissions and re-submit new 510(k) premarket notifications to FDA. This would be costly and burdensome for the Company, and there is no guarantee that the redesigned Molekule devices to meet the energy efficiency standards would be able to achieve the performance level required for a medical device.

Given the importance of maintaining access to safe and effective FDA-cleared air purification devices, the burdens associated with any design changes which could impact market availability, and the rigorous standards of FDA review prior to granting Class II medical device


Molekule
1301 Folsom St, San Francisco CA 94103
www.molekule.com

clearance, Molekule believes that an exemption from federal energy efficiency requirements for such FDA-cleared devices is necessary.

***Combined Technology Devices Adjustment***

As indicated above, a key differentiating factor of Molekule's devices, and potentially other emerging technologies in the air purifier space, is the ability to destroy pollutants such as microbes, aeroallergens and chemicals. Existing standards such as the AHAM AC-1 Protocol are solely focused on the particle removal mechanisms that are governed by the mechanical filtration efficiency and flow rate of purifiers under these tests. Based on research conducted on Molekule devices, and as one example of the limits of these standard tests, we believe these methods are particularly limited in determining the efficacy of air purifiers in removing and oxidizing aeroallergens.[2] The results of this preclinical study indicated that even at very low and undetectable levels of aeroallergens, there can still be effects on airway inflammation and lung functions. In addition, the PECO filter showed enhanced effectiveness as compared to a true HEPA rated filtration device used in the study. Clearly, air is more complex than commonly understood and the standard methods of determining the efficacy of air purifiers are limited. Any energy efficiency standards should be set to encourage, rather than dampen, the development of combined or novel technologies that focus on pollutants other than traditional particulates.

Devices that perform well against pollutant classes such as microbes and organic chemicals may require technology stacks and energy usage that are beyond what is needed for traditional mechanical filtration. Judging a purifier's effectiveness solely on particle removal efficiency without considering these other pollutant classes is an inappropriate measure of a device's energy efficiency relative to its potential benefits.

Further, many proposed and existing standards for microbes and chemicals, including proposed AHAM AC-4 and AHAM AC-5 tests and the NRCC_54013 protocol, will only gauge the initial reduction of pollutants. An important benefit of Molekule's devices is that the destruction of pollutants prevents the potential re-release of any captured pollutants. As an example, traditional carbon-based filters used to capture VOCs may have decreased performance over time as the filters become saturated or environmental changes occur that can lead to the release of previously captured VOCs. Molekule's technology allows for sustained performance over time by destroying pollutants as they move through the filter, including pulling otherwise adsorbed or captured pollutants through the PECO filter, thus reducing the saturation effect and permitting consistent performance with minimal degradation throughout the recommended life of the filter.

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[2] We have recently completed a 6-week preclinical study to evaluate efficacy of air purifiers in alleviating aeroallergen related health impacts. The animal study was completed at a research university lab.



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1301 Folsom St, San Francisco CA 94103
www.molekule.com

Due to the limitations discussed above with respect to standard particle removal tests and the complexity and limitations of existing proposed tests for microbes and chemicals, Molekule recommends that any federal energy efficiency requirements for air purifiers include an energy allowance mechanism for air purifiers utilizing combined technologies such as a fan and ultraviolet light sources that are intended to capture and destroy microbes and organic chemicals. Molekule proposes a minimum 15% allowance as compared to any energy efficiency standard that is tied to standard particle removal performance.

***Automatic/Standby Mode Adjustment***

Current and proposed energy efficiency standards or measurement methods, such as Energy Star 2.0 and AHAM AC-7 currently utilize a device's maximum running speed as the default operating mode for devices. As a manufacturer of air purifiers that include air sensors in our primary models, we know that the typical continuous use case is to operate in "Auto" mode or at a level lower than the maximum running speed. Molekule's internal data indicates that the use of Auto Mode, coupled with other common user behavior of selecting speeds lower than the maximum speed, results in more than 50% energy savings as compared to the energy usage if the device was operated continuously at maximum speed. A typical user would either use "Auto" mode or a lower running speed and utilize the maximum running speed in situations where there are higher levels of pollutants, such as strong odors, during cooking or during periodic external environmental factors such as wildfires.

The combination of "Auto" mode with particulate sensors on Molekule's Air Pro and Air Mini+ devices allow the devices to automatically increase the fan speed when higher levels of particulates are detected but otherwise operate at a slower speed to help manage lower levels of such pollutants. Energy efficiency requirements should account for the typical operation of the device rather than only the maximum speed, particularly for devices that employ sensors that facilitate automation relative to indoor air quality conditions.

"Auto" or similar modes can vary significantly among manufacturers, depending on factors such as the availability of sensor technology, software sophistication and overall device functionality. Such "Auto" modes will result in devices functioning at speed levels lower than the maximum fan speed for significant, and likely predominant, periods of use.

Due to the complexity of this area and the relatively nascent use of integrated sensor technology and associated device operation data, we believe that further input and analysis will be required to establish appropriate standards to account for "Auto" mode operation among different types of devices. However, given our findings that consumers do not operate their devices on the maximum speed for typical continuous use cases, Molekule recommends a delayed implementation of energy efficiency requirements for devices with such "automatic" or "standby" functionality until such standards are determined.



Molekule
1301 Folsom St, San Francisco CA 94103
www.molekule.com

*Conclusion*

In conclusion, the air is a complex environment that includes far more potentially harmful pollutants than simple particulate matter. Reduction of bioaerosols, organic chemicals and viruses is crucial to air quality (especially in medical settings or when necessary for medical purposes), requires testing and standards beyond traditional CADR testing with smoke particles and may be more easily addressed through emerging technologies, such as Molekule's, that have differing energy requirements than traditional air purifiers. Any federal energy efficiency requirements for air purifiers should account for these types of devices, the public health need for safe and effective FDA-regulated air purifiers, and the continuing innovation in the field. We appreciate the opportunity to provide these comments and would be happy to provide additional information if helpful.

Sincerely,

David Sanabria
Vice President of Hardware Engineering
and Technology Advancement



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